

BACKGROUND

- The 129th Boston Marathon will be run on April 21st, 2025, with approximately 30,000 athletes, including many of the world's premier marathon runners.
- The Boston Marathon course contains several elevation changes, including the infamous Heartbreak Hill during mile 21.
- The B.A.A Racing App provides in-race splits for runners using RTRT.me live data.
- The app estimates when runners will reach future checkpoints and the finish line by extrapolating their last split pace.

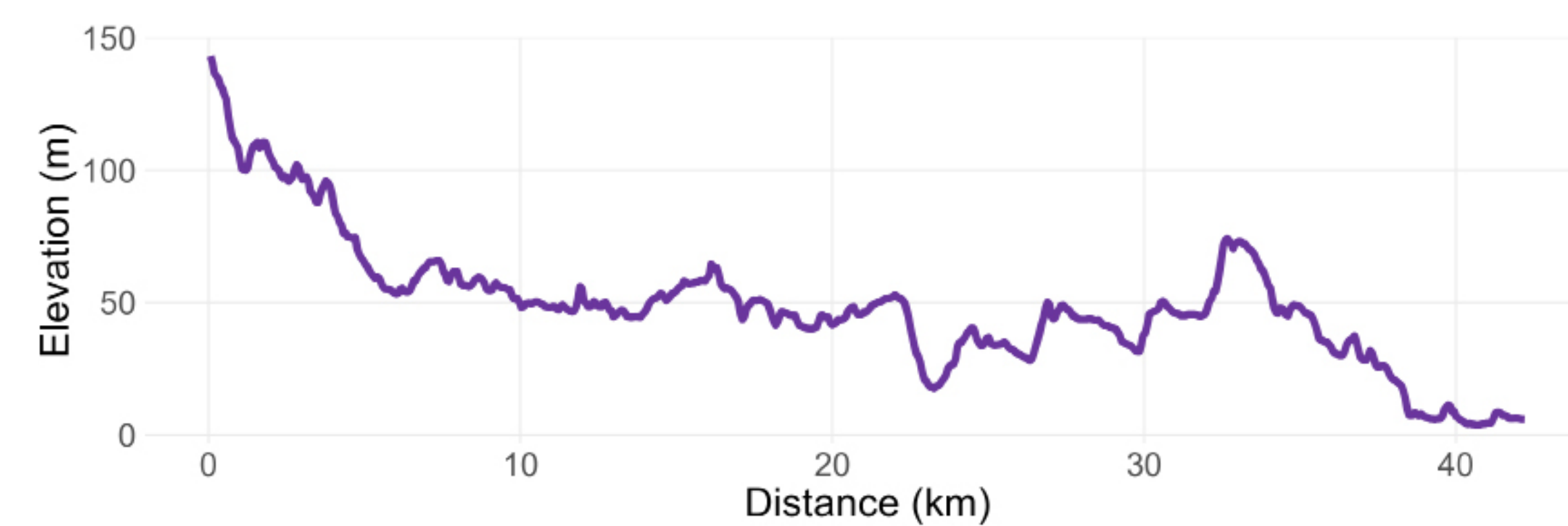


Figure 1. Boston Marathon Course Elevation

METHODS

Data Collection

- I gathered data from the 2021 to 2024 Boston Marathons using the RTRT.me API.
- In those four races, runners' times were recorded at thirteen checkpoints: 5k, 10k, 15k, 20k, 13.1mi, 25k, 30k, 20mi, 21mi, 35k, 40k, 25.2mi, and 26.2mi.
- All non-finishers or those who had missing splits were removed from the dataset.
- After cleaning, 89,448 runners were included in the complete dataset.

Statistical Approach

Modeling: Distinct random forest models were trained using the *ranger* package for each combination of checkpoints (78 total). Predicted times were computed using response output and prediction error was computed using 0.05 and 0.95 quantiles.

Random Forest Features: Age, sex, class, and all checkpoint times recorded.

Random Forest Response Variable: Future checkpoint/finish time

To assess the accuracy of this approach, the models were trained on the 2021-23 races (64,158 runners) and tested on the 2024 race (25,290).



RESULTS

Time Prediction Accuracy

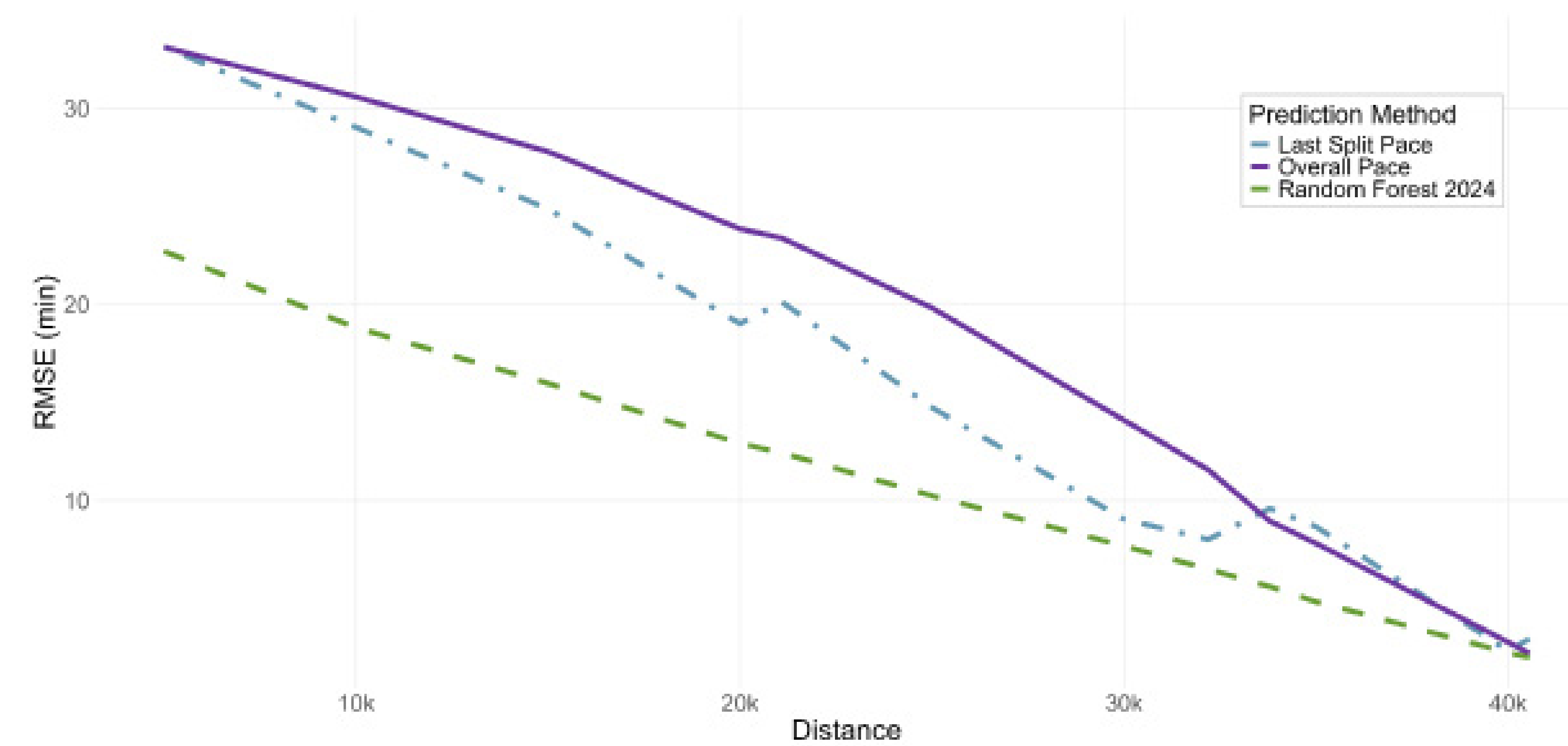


Figure 2. Random Forest Predictions for the 2024 Race had Lower Errors than the Traditional Methods

- Error is measured as prediction - actual.
- Extrapolating overall pace results in 96% of the predictions being faster than the actual finish time.
- 65% of random forest predictions are faster than the actual finish time.

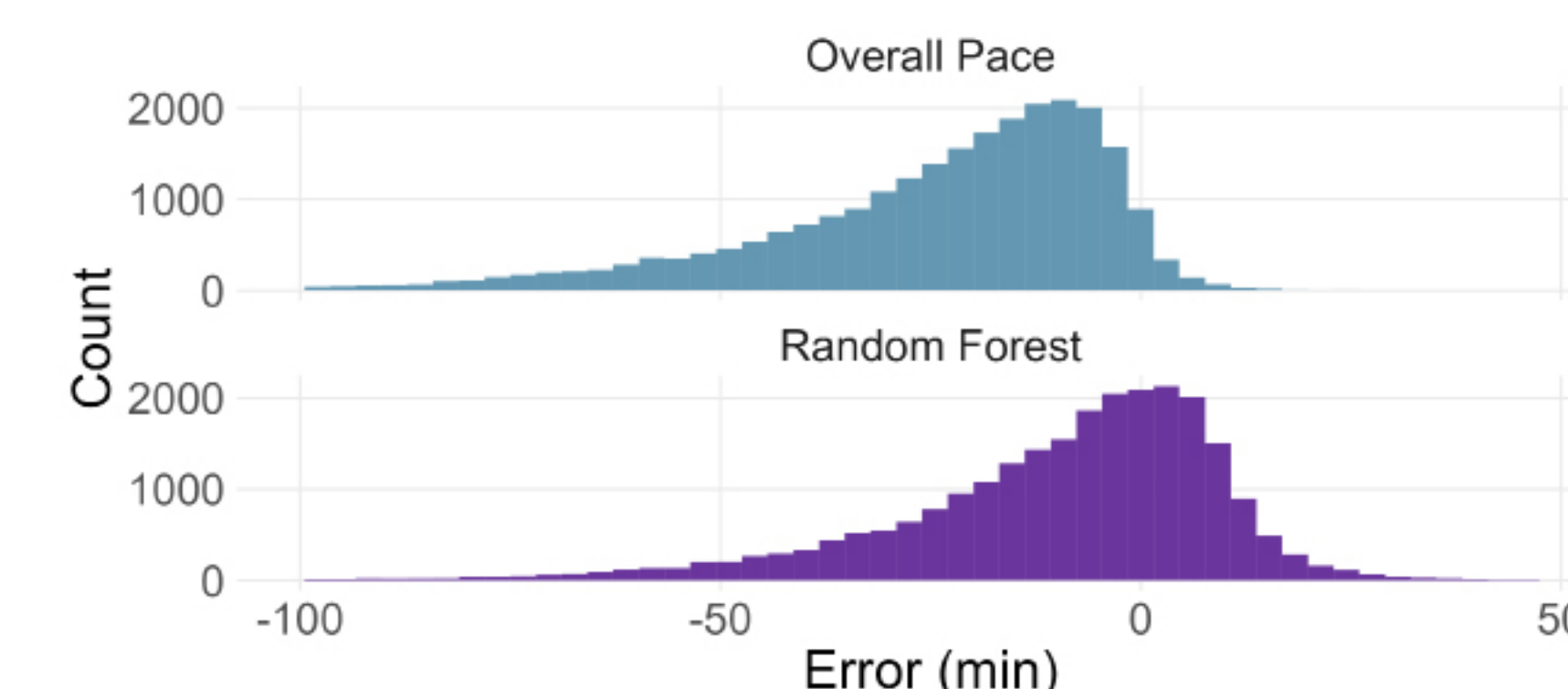


Figure 3. Histogram of Finish Time Prediction Error at 5k

Quantile Predictions

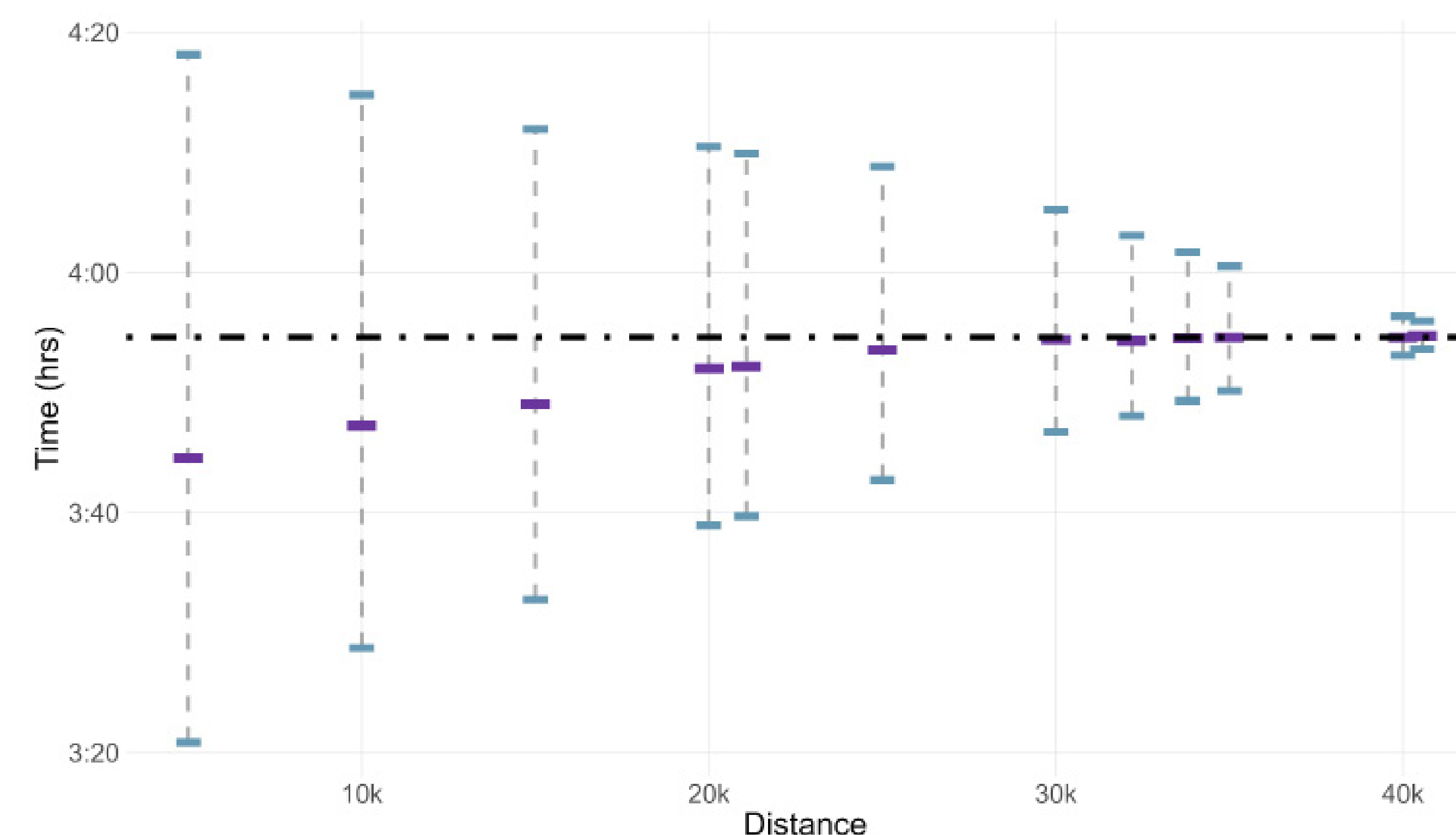


Figure 4. Average Finish Time Quantile Predictions

CONCLUSIONS

Random forest models significantly improve upon the existing prediction methods for the Boston Marathon. At some points, they reduce the prediction error of existing methods by nearly 50%. Prediction quantiles also provide information about the uncertainty of time predictions, which is not currently available on the Boston Marathon app's live results.

Significance

More accurate in-race predictions can:

- Improve analysis of the professional competition during the race.
- Help runners' friends and family track them throughout the race and see them as much as possible.
- Give runners a better understanding of how they can achieve their marathon time goals.

NEXT STEPS

- Create an app to deliver improved predictions in real time.
- Assess the accuracy of these models on the 2025 Boston Marathon.
- Incorporate runners' previous marathon times into the predictions.
- Include weather conversions to account for the extreme weather conditions that sometimes occur (snow, driving rain, extreme heat, etc.).
- Quantify the effects of new advancements in racing technology like "super shoes" and gels.

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Acknowledgements: Thanks to Dr. Thomas Metzger for giving his time and statistical guidance to this project.